

FOUNDATION COURSE

REV2021

Business Statistics

Course Code: 2141 | Semester: 2 | Credits: 3

[\[Download Official Syllabus PDF\]](#)

Course Overview

Teaching Scheme

- **Lecture (L):** 2 Hours/Week
- **Tutorial (T):** 0 Hours/Week
- **Practical (P):** 2 Hours/Week
- **Total Periods:** 60 per Semester

Course Objectives

- Thorough knowledge in basic concepts of statistics, classification, and tabulation.
- Skill in applying statistical tools to interpret business problems.
- Understanding graphical representation and data analysis.

Course Outcomes (COs)

CO No.	Description	Duration	Cognitive Level
CO1	Explain basic concepts, classification, tabulation, and presentation of data.	13 Hours	Applying
CO2	Apply measures of central value and dispersion.	19 Hours	Applying
CO3	Create relationship between variables by means of correlation and regression.	12 Hours	Applying
CO4	Construct and compare index numbers.	14 Hours	Applying

Module 1: Basic Concepts of Statistics

Duration: 13 Hours | CO1

1.1 Meaning and Functions of Statistics

Statistics is the science of collecting, organizing, analyzing, interpreting, and presenting data. In a business context, it provides a numerical basis for decision-making.

Functions of Statistics:

- Simplification:** It simplifies complex data into understandable forms (averages, percentages).
- Comparison:** It helps in comparing data across different time periods or regions.
- Policy Formulation:** Helps governments and businesses frame policies based on trends.
- Forecasting:** Predicting future trends based on past data.

Malayalam Note: സ്റ്റാറ്റിസ്റ്റിക്സ് എന്നത് വിവിധ വിധത്തിൽ ക്രമീകരിച്ച, സംഗ്രഹിച്ച, വിശ്ലേഷിച്ച, വ്യാഖ്യാനിച്ച, പ്രദർശിപ്പിച്ച ഡാറ്റയെക്കുറിച്ചാണ്. ഇത് വ്യക്തമാക്കുന്നു, താരതമ്യം ചെയ്യുന്നു, തത്വം നിർണ്ണയിക്കുന്നു, പ്രവണതകൾ കണ്ടെത്തുന്നു. സ്റ്റാറ്റിസ്റ്റിക്സ് വ്യക്തമാക്കുന്നു, താരതമ്യം ചെയ്യുന്നു, തത്വം നിർണ്ണയിക്കുന്നു, പ്രവണതകൾ കണ്ടെത്തുന്നു.

1.2 Types of Data and Collection Methods

Primary Data

Data collected for the first time by the investigator for a specific purpose.

- Direct Personal Interview
- Mailed Questionnaires
- Telephone Interviews

Secondary Data

Data which has already been collected by someone else and processed.

- Government Publications (Census)
- Journals and Reports
- Websites and Newspapers

1.3 Classification and Tabulation

Classification: The process of arranging data into groups or classes according to common characteristics.

- Geographical (Area-wise)
- Chronological (Time-wise)
- Qualitative (Attributes like gender, literacy)
- Quantitative (Numerical values like height, weight)

Tabulation: Logical and systematic arrangement of statistical data in columns and rows.

1.4 Graphical Representation

Visual representation helps in quick understanding of data distribution.

- **Histogram:** A set of rectangles with bases as class intervals and heights proportional to frequencies.
- **Frequency Polygon:** Formed by joining the mid-points of the tops of the rectangles in a histogram.
- **Ogive (Cumulative Frequency Curve):** Used to find the median. There are two types: "Less than" and "More than".

Module 2: Measures of Central Value & Dispersion

2.1 Measures of Central Value (Averages)

An average is a single value that represents the entire set of data.

Arithmetic Mean

Sum of all observations divided by the number of observations.

$$\text{Mean } (\bar{x}) = \Sigma x / n$$

Median

The middle value of the data when arranged in ascending or descending order.

Mode

The value that occurs most frequently in the data set.

2.2 Measures of Dispersion

Dispersion measures the extent to which data items are scattered around the central value.

Standard Deviation (σ)

The square root of the arithmetic mean of the squares of deviations from the arithmetic mean.

$$\sigma = \sqrt{[\Sigma (x - \bar{x})^2 / n]}$$

Example: Calculate Mean and SD for 2, 4, 6.

Mean: $(2+4+6)/3 = 4$.

Deviations ($x-\bar{x}$): -2, 0, 2.

Squares: 4, 0, 4. Sum = 8.

SD: $\sqrt{(8/3)} \approx 1.63$.

Module 3: Correlation and Regression Analysis

Duration: 12 Hours | CO3

3.1 Correlation Analysis

Correlation studies the relationship between two variables. If change in one variable results in a change in the other, they are correlated.

- **Positive Correlation:** Both move in the same direction (e.g., Price and Supply).
- **Negative Correlation:** Move in opposite directions (e.g., Price and Demand).

Karl Pearson's Coefficient of Correlation (r)

$$r = \frac{\sum xy}{\sqrt{(\sum x^2 * \sum y^2)}}$$

Where $x = (X - \bar{X})$ and $y = (Y - \bar{Y})$.

Malayalam Note: കോർറിലേഷൻ കോഫിഷ്യൻ്റ് രണ്ടു വേരിയബിളുകളിനുമിടയിലെ ബന്ധം അളക്കുന്നു. 'r' ന്റെ മൂല്യം -1 ന്റെ +1 ന്റെ ഇടയിലായിരിക്കും.

3.2 Regression Analysis

Regression is used to estimate the value of one variable (dependent) based on the value of another (independent).

- **Regression Line of Y on X:** $Y - \bar{Y} = b_{yx}(X - \bar{X})$
- **Regression Line of X on Y:** $X - \bar{X} = b_{xy}(Y - \bar{Y})$

Module 4: Index Numbers

Duration: 14 Hours | CO4

4.1 Construction of Index Numbers

Index numbers are specialized averages used to measure changes in a variable (like price) over time.

Laspeyres' Index

Uses base year quantities (q_0) as weights.

$$P_{01} = [\Sigma p_1 q_0 / \Sigma p_0 q_0] \times 100$$

Paasche's Index

Uses current year quantities (q_1) as weights.

$$P_{01} = [\Sigma p_1 q_1 / \Sigma p_0 q_1] \times 100$$

Fisher's Ideal Index

Geometric mean of Laspeyres and Paasche index.

$$\text{Fisher} = \sqrt{(\text{Laspeyres} \times \text{Paasche})}$$

4.2 Tests of Adequacy

- **Time Reversal Test:** $P_{01} \times P_{10} = 1$
- **Factor Reversal Test:** $P_{01} \times Q_{01} = \Sigma p_1 q_1 / \Sigma p_0 q_0$

Fisher's Index satisfies both tests, hence it is called "Ideal".

Formula Bank

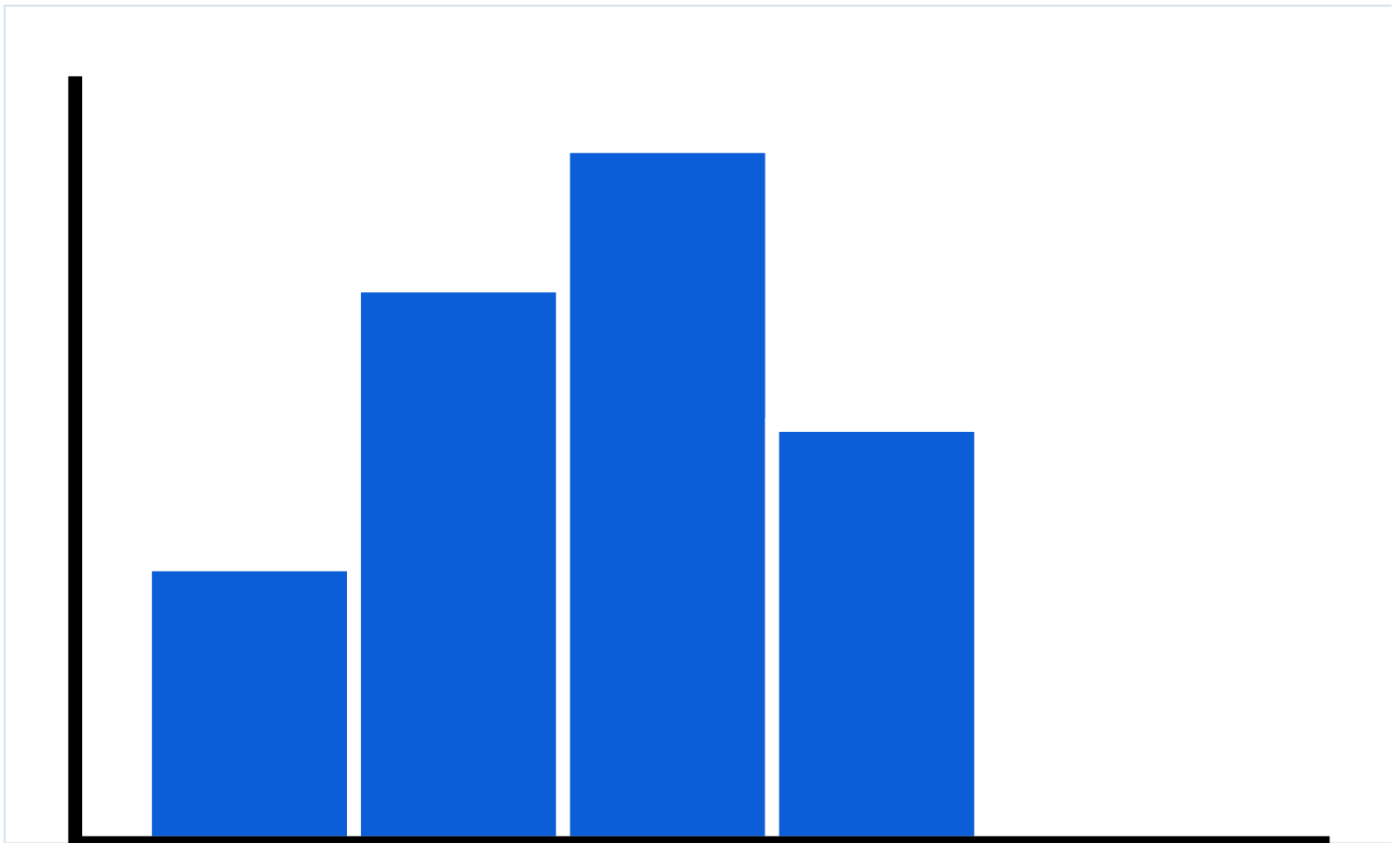
Central Tendency

- Mean: $\Sigma x / n$
- Weighted Mean: $\Sigma wx / \Sigma w$
- Combined Mean: $(n_1\bar{x}_1 + n_2\bar{x}_2) / (n_1 + n_2)$

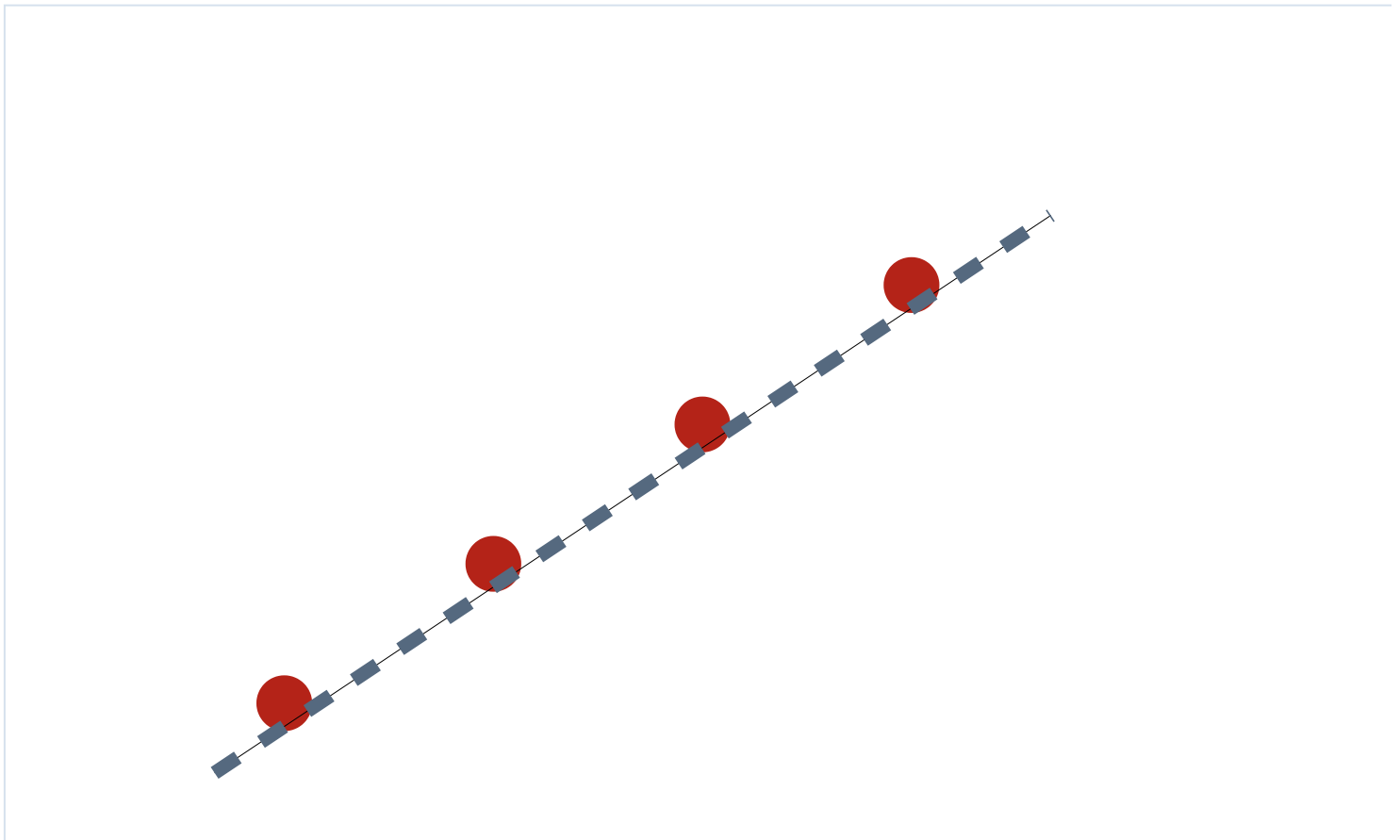
Dispersion & Correlation

- Standard Deviation: $\sqrt{[\Sigma(x-\bar{x})^2 / n]}$
- Karl Pearson r: $\Sigma xy / \sqrt{(\Sigma x^2 * \Sigma y^2)}$
- Spearman's Rank: $1 - [6\Sigma D^2 / n(n^2-1)]$

Visual Library



Basic Histogram Representation



Positive Correlation Scatter Plot

Module-wise Practice Questions

Q1: Distinguish between Primary and Secondary data. (5 Marks)

Primary data is original data collected for the first time, while secondary data is already existing data collected by others. Primary data is more accurate but expensive; secondary data is cheaper but may not fit the specific needs perfectly.

Q2: What are the requisites of a good average? (3 Marks)

1. Rigidly defined. 2. Easy to understand and calculate. 3. Based on all observations. 4. Least affected by fluctuations of sampling.

Q3: Explain the properties of Karl Pearson's coefficient of correlation. (5 Marks)

1. The value of 'r' lies between -1 and +1. 2. It is independent of change of origin and scale. 3. If $r = +1$, it is perfect positive correlation. 4. If $r = -1$, it is perfect negative correlation. 5. If $r = 0$, there is no linear correlation.

Practice Model Paper

Section A (Answer all questions - 2 Marks each)

1. Define Statistics.
2. What is an Ogive?
3. State the formula for Weighted Mean.
4. Define Index Number.

Section B (Answer any 4 - 5 Marks each)

1. Explain the limitations of Statistics.
2. Calculate Arithmetic Mean: 10, 20, 30, 40, 50.
3. Distinguish between Correlation and Regression.
4. Explain Time Reversal Test.

Section C (Answer any 2 - 10 Marks each)

1. Explain different methods of collecting primary data.
2. Calculate Standard Deviation for the following data: 5, 10, 15, 20, 25.
3. Construct Fisher's Ideal Index from the given price-quantity data.

Quick Revision

- **Statistics:** Collection, Analysis, Interpretation.
- **Mean:** Average; **Median:** Middle; **Mode:** Most frequent.
- **Standard Deviation:** Best measure of dispersion.
- **Correlation (r):** Relationship strength (-1 to +1).
- **Fisher's Index:** Ideal index (satisfies TRT and FRT).

Glossary

Data

Facts or figures from which conclusions can be drawn.

Variable

A characteristic that varies from one item to another.

Population

The entire group under study.

Sample

A part of the population selected for study.

Official Source Declaration

This lesson is based on the official syllabus for **2141 — Business Statistics (Revision 2021)** provided by the State Institute of Technical Teachers' Training & Research (SITTTR), Kerala.

Source URL: [SITTTR Kerala - Course 2141](#)

മലയാളം പിന്തുണ (Malayalam Support)

പ്രധാന ആശയങ്ങൾ:

- സമാപ്തത (Mean):** ഒരു കൂട്ടം ഡാറ്റയുടെ സമാപ്തത കണക്കാക്കുന്നത് എങ്ങനെ.
- മധ്യസ്ഥത (Median):** ഒരു കൂട്ടം ഡാറ്റയുടെ മധ്യസ്ഥത കണക്കാക്കുന്നത് എങ്ങനെ.
- രീതി (Mode):** ഒരു കൂട്ടം ഡാറ്റയുടെ ഏറ്റവും കൂടുതൽ ഉണ്ടാകുന്ന രീതി കണ്ടെത്തുന്നത് എങ്ങനെ.
- സംബന്ധം (Correlation):** രണ്ട് വേരിയബിളുകൾ തമ്മിലുള്ള ബന്ധം അളക്കുന്നത് എങ്ങനെ. പോസിറ്റീവ്, നെഗറ്റീവ്, ശून്യ.